

M-Link Data Layer

M-link data layer operates with words of 4 bytes width.

Data Frame Format

Sync	Type	Length	Seq	Dst	Src	Data	CRC
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field	offset	size	description
Sync	0	2	Sync word. 0x2A, 0x50 = “*P”
Type	2	2	Frame type
Length	4	2	Frame length in 32-bit words. Valid length is 4 – 511 (data size 0-507)
Seq	6	2	Frame sequence number (counter)
Dst	8	2	Destination Address
Src	10	2	Source Address
Data	12	0-2028	Payload data. 0-507 32-bit words
CRC	last	4	CRC32

Field offset and size in bytes.

Data flow: for each endpoint, SEND is what endpoint is sending

Data Frame Types

type	description	data format
0x0100	Run mode data	application specific
0x0101	Register R/W request	Frame_101_102
0x0102	Register R/W response	
0x0103	Status response	application specific
0x0105	RAM R/W request	Frame_105_106
0x0106	RAM R/W response	
0x0107	Configuration EEPROM R/W request	Frame_107_108

0x0108	Configuration EEPROM R/W response	
0x0110	FVME run mode data	VmeDaqRawdataFormat
0x0111	FVME request	Frame_111_112
0x0112	FVME response	
0x1001	JTAG single op request	Frame 1001-1004
0x1002	JTAG single op reply	
0x1003	JTAG shift 32 request	
0x1004	JTAG shift 32 reply	
0x5A5A	Loopback Test	any data
0x5354	Stream data	MStream

Reserved Addresses

address	description
0xFEFE	point-to-point
0xFDFD	PCIe carrier board

MLink